Knowledge post-test

Total score 20 points

Participant ID: _____

Part I. Multiple-Choice Questions (8 questions, 1 point each, total 8 points)

1. In a ramp merging area, which factor most directly affects whether a merging vehicle can successfully enter the mainline?

A. Number of mainline lanes
B. Available gaps and their duration
C. Average speed of mainline traffic
D. Ramp design length

2. Which of the following situations is most likely to cause a sharp deceleration conflict in a ramp merging area?

A. The merging vehicle completes speed matching in advance and merges smoothly
B. The merging vehicle forces entry at the end of the acceleration lane
C. Mainline vehicles maintain large headways
D. The speed distribution of the mainline traffic is uniform

3. One important reason why traffic conflicts are significantly higher in weaving sections than in ordinary road segments is that:

A. Vehicles only perform longitudinal movements
B. There is no path competition between vehicles
C. Multiple vehicles have interacting lane-changing demands within a limited space
D. There is no speed variation in the segment

4. When analyzing a ramp merging scenario, which of the following is most appropriate for identifying an increase in risk?

A. A decrease in mainline traffic density
B. An increase in speed differences combined with a reduction in acceptable gaps
C. No change in the ramp entrance location
D. An increase in the number of lanes

5. In a high-density weaving section, which behavior is most likely to exacerbate traffic conflicts?

A. Completing the desired lane change before entering the weaving section
B. Dynamically adjusting lane-changing timing based on surrounding traffic conditions
C. Performing lane changes only when approaching the target position
D. Maintaining the same speed as the leading vehicle with reduced speed fluctuations

6. From a traffic engineering perspective, the complexity of a weaving section mainly lies in:

A. Differences in vehicle operating speeds
B. Spatial path competition and crossing interactions among vehicles
C. Temporal fluctuations in traffic flow
D. Mixed traffic composed of different vehicle types

7. If the instructional goal is to help students understand the mechanisms of traffic conflict formation, which of the following learning activities is most effective?

A. Summarizing typical conflict scenarios through case studies
B. Analyzing vehicle interactions based on dynamic traffic scenarios
C. Systematically reviewing definitions of traffic conflict-related terms
D. Learning basic design parameters of different road types

8. Which of the following best reflects a student's ability to transfer knowledge of ramp merging?

A. Accurately describing the basic definitions and characteristics of ramp merging
B. Explaining how conflicts vary under different traffic conditions
C. Applying conflict analysis methods from merging scenarios to closely spaced entrance–exit interactions
D. Reproducing example cases presented in class

Part II. True/False Questions (4 questions, 1 point each, total 4 points)

9. In a ramp merging area, traffic conflicts will not occur as long as merging vehicles accelerate sufficiently.

[ ] True [ ] False

10. Traffic conflicts in weaving sections are related not only to individual vehicle behavior but also to multi-vehicle path competition.

[ ] True [ ] False

11. An increase in traffic density typically compresses the available operational space for lane changing and merging.

[ ] True [ ] False

12. If a student can identify where a conflict occurs, it means the student has fully understood the mechanism of conflict formation.

[ ] True [ ] False

Part III. Short Answer Questions (3 questions, total 8 points)

13. Explain why "acceptable gap" and "speed coordination" are two key concepts in ramp merging areas. (3 points)

14. Compare the differences in traffic conflict formation mechanisms between standard ramp merging sections and weaving sections. (3 points)

15. In a road segment where an entrance and an exit are located close to each other, entering vehicles and exiting vehicles adjust lanes simultaneously within a short distance. From the perspective of traffic conflicts, briefly analyze the main sources of risk in this scenario. (2 points)

Answer key and scoring criteria

Answer key B B C B C B B C False True True False

Question 13

Reference points

Acceptable gap determines whether merging can be completed safely

Speed coordination affects how well the merging vehicle matches mainline traffic

Both jointly influence whether conflicts occur

Scoring criteria

0 points: Incorrect or blank

1 point: Mentions only one concept

2 points: Mentions both concepts but explanation is limited

3 points: Clearly explains both concepts and their combined effect

Question 14

Reference points

Merging sections mainly involve one-directional entry into the mainline

Weaving sections involve bidirectional or multi-directional lane changes

Weaving sections feature more compact space, stronger interactions, and more pronounced path competition

Scoring criteria

0 points: Incorrect or blank

1 point: Only states that weaving sections are “more complex”

2 points: Identifies one or two differences

3 points: Provides a structured comparison including interaction direction, spatial constraints, and path competition

Question 15

Reference points

Multiple vehicles perform lane changes within a short distance

Conflicts arise due to overlapping target paths of entering and exiting vehicles

Speed differences, limited space, and late lane changes increase risk

Scoring criteria

0 points: Invalid or irrelevant answer

1 point: Identifies part of the risk sources

2 points: Identifies multiple key risk sources with basic reasoning